

An Interpretability and Distillation Study of LGATr (Lorentz Geometric Algebra Transformer)

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Abstract

We study the interpretability of the Lorentz Geometric Algebra Transformer (LGATr), a Lorentz-equivariant state-of-the-art model for jet tagging that leverages Geometric Algebra to model particle interactions over a 4-D space-time. While interpretability research has matured for standard transformers, the inner workings of Lorentz-equivariant models remain largely unexplored.

We also investigate whether Lorentz equivariance can be transferred from an equivariant teacher model to a non-equivariant student using physics-aware knowledge distillation. This approach offers a potential solution to the high latency of equivariant architectures, which often limits their deployment in real-time edge applications like Large Hadron Collider (LHC) hardware triggers.

We train LGATr on the TopTagging dataset and visualize the attention heatmaps. Our analysis reveals that while LGATr’s attention heatmaps lack clear visual geometric groupings, its attention score distribution closely mirrors that of the non-equivariant Particle Transformer (ParT), exhibiting a statistically sparse and highly polarized pattern. Interestingly, our knowledge distillation experiments reveal that a Multi-Layer Perceptron (MLP) student trained from scratch outperforms the distilled student. We hypothesize that this drop in distillation efficacy is due to the pipeline’s reliance on raw Cartesian coordinates, unlike prior work that utilized engineered kinematic features, highlighting a critical bottleneck in distilling geometric-algebra-based representations.

1 Introduction

Machine learning algorithms have become indispensable for analyzing the massive volumes of data generated by particle physics experiments, such as ATLAS and CMS at the CERN Large Hadron Collider (LHC). Among these tasks, jet

tagging—the process of identifying the parent particle that initiates a collimated spray of outgoing particles—is critical for both Standard Model measurements and searches for new physics (Qu and Gouskos, 2020; Kasieczka et al., 2019a). Attention-based transformer models have recently revolutionized this domain by effectively capturing intricate correlations between particles within a jet. These architectures, particularly those that incorporate physical symmetries directly into the network weights (Gong et al., 2022; Spinner et al., 2024; Qu et al., 2024), have pushed the boundaries of classification accuracy, transforming the data analysis workflow in high-energy physics.

Despite their performance, these models often function as “black boxes,” making it difficult to verify if they are learning true physical laws or merely latching onto spurious data correlations. Furthermore, while models with strong inductive biases, such as Lorentz equivariance, offer superior accuracy, they are computationally expensive and memory-intensive (Petitjean et al., 2025). This poses a major challenge for hardware trigger systems at the LHC (such as the Level-1 trigger), which operate under strict power and resource constraints and require inference latencies on the order of a single microsecond (Liu et al., 2023). To be useful in real-time “online” applications, a model must not only be accurate but also extremely fast and physically grounded, creating a tension between architectural complexity and operational efficiency.

Recent research has seen a surge in architectural innovation, ranging from general-purpose Graph Neural Networks (Gong et al., 2022) to custom models like the Particle Transformer (ParT) (Qu et al., 2024). However, while mechanistic interpretability has matured significantly in the context of Natural Language Processing, its application to equivariant physics models remains largely unexplored. We lack a clear understanding of

how internal activations in these models construct physical representations (Wang et al., 2024) (Legge et al., 2025). Additionally, while Knowledge Distillation (KD) has been used to compress large models in other fields, its ability to specifically transfer physical understanding—such as Lorentz equivariance—from a complex teacher to a simplified student for edge deployment has not been sufficiently demonstrated, especially when operating on raw 4-momenta (Liu et al., 2023).

In this work, we investigate the Lorentz Geometric Algebra Transformer (LGATr), the current state-of-the-art physics-aware model for jet tagging, to bridge these gaps. We propose a dual-track study: first, we apply mechanistic interpretability techniques to LGATr’s attention mechanisms to understand how the model builds a physics-aware internal logic. Second, we explore a physics-aware knowledge distillation framework. By using LGATr as a teacher model, we attempt to distill its inherent Lorentz equivariance into a non-equivariant, high-speed MLP student.

We conduct our experiments using the standard TopTagging dataset (Kasieczka et al., 2019b), establishing LGATr as a robust performance baseline. Our interpretability results demonstrate that LGATr exhibits visually complex but statistically sparse attention whose score distributions closely align with findings from non-equivariant models like ParT. Conversely, our distillation experiments yield an unexpected result: the baseline MLP trained from scratch outperforms the KD-trained student. We hypothesize that the direct use of Cartesian coordinates in the LGATr pipeline, without the specialized coordinate transformations used in previous distillation studies, hinders the effective transfer of Lorentz equivariance to the linear student model.

Contributions. Our main contributions are as follows:

- **Mechanistic Interpretability of Equivariance:** We conduct a granular analysis of LGATr’s internal attention mechanisms, demonstrating that its attention score distributions mirror the sparse behavior observed in standard ParT models, despite lacking clear visual structure in the heatmaps.
- **Distillation Bottleneck Identification:** We investigate a physics-aware distillation framework and identify a critical limitation: relying

on raw Cartesian inputs prevents simple non-equivariant students from effectively mimicking complex Geometric Algebra teachers.

- **Open Source Implementation:** We provide our codebase, including the full LGATr interpretability pipeline and the KD framework, to support reproducibility and further research.

2 Related Work

2.1 Transformer Architectures and Interpretability in Jet Tagging

The adoption of Transformer architectures for jet tagging was popularized by the Particle Transformer (ParT), a non-equivariant model that introduced physics-inspired pairwise interactions to augment standard attention mechanisms (Wang et al., 2024). ParT explicitly injects physical observables (like ΔR and mass) into the self-attention calculation. Further investigations explored why attention in ParT is naturally sparse, revealing that the model focuses on a small subset of “hard” interactions that define jet substructure and that its attention score distributions follow highly specific patterns, practically acting as a hard clustering mechanism (Legge et al., 2025).

Gap addressed by this paper: Existing studies have largely relied on visualizing attention for non-equivariant models that require explicit physical biases to be injected. We extend this inquiry to LGATr, demonstrating that despite the mathematical complexity of Geometric Algebra operating on raw Cartesian coordinates, the model internally arrives at similarly sparse, physically meaningful representations as ParT.

2.2 Efficiency and Knowledge Distillation for LHC Triggers

Deep learning models at the LHC face significant deployment challenges. The Level-1 trigger relies on Field Programmable Gate Arrays (FPGAs) to make decisions in under a microsecond. Scaling $O(N^2)$ transformer models to this latency is currently prohibitive. To bridge the gap, prior work utilized Knowledge Distillation (KD), demonstrating that a LorentzNet teacher could successfully transfer knowledge to simpler MLP student models (Liu et al., 2023). A key factor in their success was explicitly converting raw inputs into engineered physical features (like relative p_T and $\Delta\eta, \Delta\phi$) before feeding them to the student.

Gap addressed by this paper: We study the distillation of the transformer-based LGATr using raw Cartesian coordinates natively, identifying the exact limits of knowledge transfer when manual feature engineering is omitted from the student’s inputs.

2.3 Lorentz-Equivariant Geometric Algebra Models

The Lorentz Geometric Algebra Transformer (LGATr) represents the current state-of-the-art, utilizing Geometric Algebra to ensure strict Lorentz equivariance (Spinner et al., 2024). Unlike earlier equivariant models that required manual scalar feature engineering, LGATr operates on the geometric properties of Cartesian 4-momenta directly by representing them as elements in a Clifford Algebra. A subsequent study introduced “LGATr_slim,” an ablated version of the network that restricts geometric operations for computational efficiency (Petitjean et al., 2025).

Gap addressed by this paper: We provide a dual-track study of LGATr that addresses both its internal transparency (comparing the full and slim variants) and the practical difficulties of distilling its geometric representations for real-time environments.

3 Methodology

3.1 Overview

Our methodology consists of two parallel experimental tracks. Figure 1 illustrates the end-to-end pipeline. First, we establish an interpretability pipeline to extract and analyze the attention mechanisms of the full LGATr against a stripped-down variant (LGATr_slim). Second, we develop a Knowledge Distillation framework to compress the representations of the LGATr teacher into a lightweight MLP student.

3.2 Geometric Algebra Embeddings and Data Representation

The input to our models consists of jet constituents. For the LGATr architectures, the input is raw Cartesian 4-momenta $p = (E, p_x, p_y, p_z)$. Instead of transforming these into traditional high-energy physics coordinates, they are explicitly embedded into Geometric Algebra multivectors within the Clifford Algebra $\mathcal{Cl}(1, 3)$. A multivector M in this space consists of 16 components:

$$M = s + v + B + t + p \quad (1)$$

where s is a scalar, v a vector, B a bivector, t a trivector, and p a pseudoscalar. LGATr operates strictly through geometric products, ensuring that any Lorentz transformation applied to the inputs naturally commutes with the operations of the network. (Spinner et al., 2024)

In contrast, for the LGATr_slim variant, the inputs and operations are restricted to decoupled scalar and 4-vector components, significantly reducing the parameter count by avoiding the full 16-dimensional multivector expansions. (Petitjean et al., 2025)

Crucially, in our distillation pipeline, we feed the exact same raw Cartesian coordinates to the MLP student. This is designed to test whether the soft labels from the teacher alone can force a non-equivariant, linear network to learn relativistic invariants without manual feature engineering.

3.3 Mechanistic Interpretability and Comparative Analysis

To interpret the model’s decision-making, we probe the self-attention layers of the transformers. In LGATr, attention weights are computed over geometric representations via a tailored dot-product in Minkowski space. (Spinner et al., 2024) We conduct a comparative analysis between the full LGATr model (12 blocks) and the LGATr_slim variant by globally aggregating attention scores across all layers and heads over batched event samples. By analyzing the sparsity and density of the attention maps, we assess how the model parses jet topology.

3.4 Physics-Aware Knowledge Distillation Framework

To address latency constraints, we utilize a Knowledge Distillation framework. The goal is to transfer the learned representations of the complex teacher T to the lightweight student S . The training objective is a linear combination of the standard cross-entropy loss (using the true hard labels y) and the Kullback-Leibler (KL) divergence between the softened output logits of the student (z_s) and the teacher (z_t):

$$\mathcal{L}_{\text{KD}} = \alpha \mathcal{H}(y, \sigma(z_s)) + (1 - \alpha) \tau^2 \mathcal{D}_{\text{KL}} \left(\sigma \left(\frac{z_s}{\tau} \right) \parallel \sigma \left(\frac{z_t}{\tau} \right) \right) \quad (2)$$

where σ is the softmax function, τ is the temperature parameter controlling the softness of the dis-

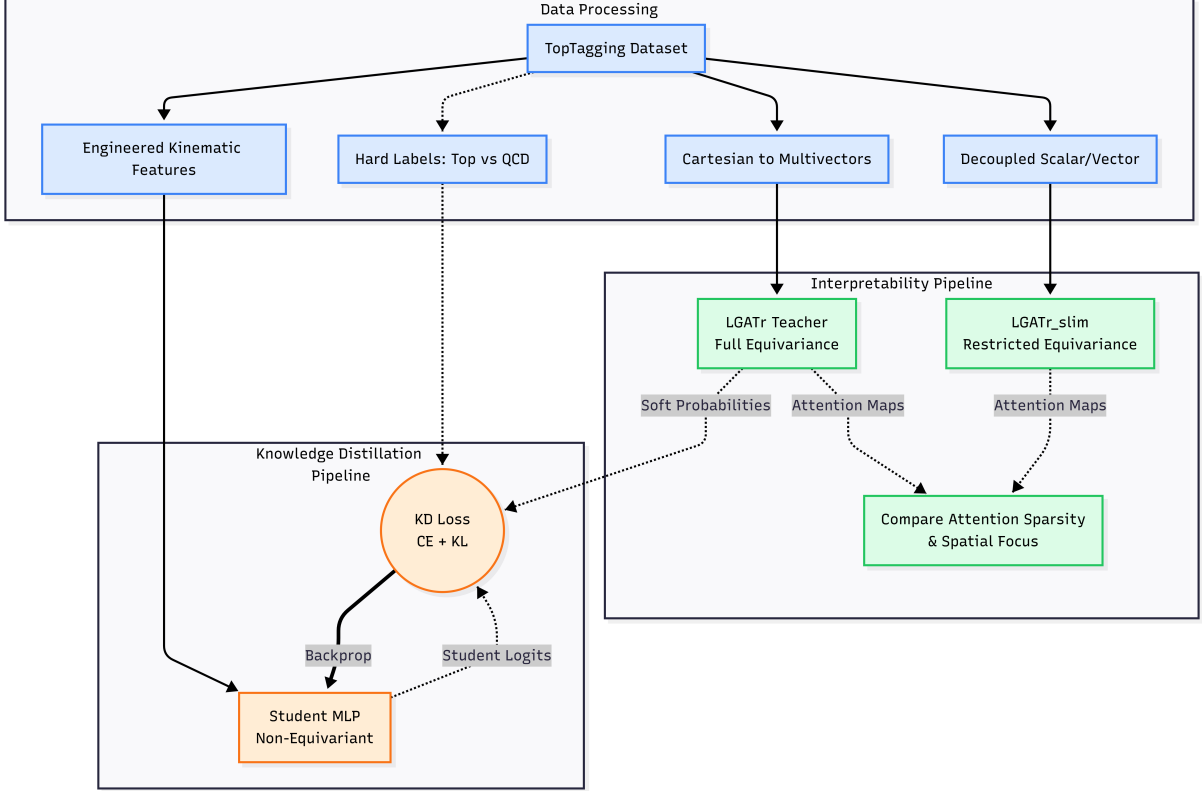


Figure 1: **Overview of the Interpretability and Distillation Pipeline.** (1) The *Interpretability Pipeline* extracts attention distributions from both the full LGATr and LGATr_slim models. (2) The *Knowledge Distillation Pipeline* attempts to train a student MLP using a combination of hard labels and soft teacher probabilities operating on raw Cartesian coordinates.

tributions, and α balances the two losses. The temperature scaling allows the student to learn from the “dark knowledge”—the secondary probabilities assigned by the teacher to incorrect classes, which encode similarities between different decay kinematics.

4 Experimental Setup

4.1 Datasets

We conduct our experiments on the standard TopTagging reference dataset (Kasieczka et al., 2019a). This dataset is a widely used benchmark for evaluating jet tagging algorithms. It consists of hadronic top-quark jets (signal) and QCD di-jet events (background) simulated at a center-of-mass energy of $\sqrt{s} = 14$ TeV using Pythia8 and Delphes.

Each jet is represented by up to 200 constituent particles. For our MLP processing pipeline, sequences are zero-padded or truncated to a fixed maximum of 128 particles per jet to accommodate strict tensor dimensions. Table 1 summarizes the dataset scale.

Dataset Split	Domain	# Events
Training	Top vs. QCD	1,200,000
Validation	Top vs. QCD	400,000
Test	Top vs. QCD	400,000

Table 1: Summary of the TopTagging dataset splits used for training, validation, and testing.

4.2 Architectures

Our experimental pipeline involves three distinct architectures. Table 2 provides a high-level overview.

- **LGATr (Teacher):** The full Lorentz Geometric Algebra Transformer (Spinner et al., 2024). The model consists of 12 transformer blocks, 16 hidden multivector channels, 32 hidden scalar channels, and 8 attention heads, adhering precisely to the hyperparameter configuration of the original study.
- **LGATr_slim:** An ablated variant of LGATr (Petitjean et al., 2025) that removes the full multivector embeddings in favor of independent

scalar and 4-vector propagation. It consists of 12 blocks, 32 vector channels, 96 scalar channels, and 8 attention heads. Notably, to compensate for the loss of higher-grade geometric representations, this variant uses wider hidden dimensions than the base model, resulting in a higher overall parameter count despite the structural ablation.

- **MLP (Student):** A standard Multi-Layer Perceptron trained to classify jets based on flattened Cartesian features (4×128 input dimensions). The network consists of 3 hidden layers with a dimensionality of 512, utilizing LeakyReLU activations and batch normalization.

Model	Params	Role
LGATr	~ 1.1 M	Teacher / Interp
LGATr_slim	~ 2.0 M	Interp baseline
MLP	~ 2.0 M	Student (KD / Scratch)

Table 2: Overview of models evaluated in this study.

4.3 Implementation and Training Details

All models are implemented in PyTorch. For the LGATr and LGATr_slim interpretation, attention maps are extracted during a forward pass over the test set using a custom evaluation script hooked to the transformer blocks.

For the knowledge distillation track, we compare the KD-trained MLP against an identical MLP trained entirely from scratch using only hard labels. Both models were trained in a distributed data-parallel environment across 4 GPUs. We employed the AdamW optimizer with a learning rate of 8×10^{-4} and a global batch size of 256. The student models were trained for a total of 100 epochs to ensure full convergence.

Performance is primarily evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC) and Background Rejection ($1/\epsilon_B$) measured at a fixed signal efficiency of $\epsilon_S = 30\%$, a critical metric for hardware triggers.

5 Results and Discussion

5.1 Interpretability of Equivariant Attention

Objective. We aim to understand how Lorentz-equivariant inductive biases manifest in the attention matrices of LGATr. Given its unique Geometric Algebra foundation, we evaluate if it be-

haves differently than standard non-equivariant transformers.

Findings. Analyzing the extracted attention heatmaps from LGATr on the TopTagging dataset, we observe a distinct phenomenon: while the attention matrices contain many zero values indicating sparsity, the heatmaps themselves appear visually messy and unstructured (see Figure 2, Left). This contrasts with the binary attention heatmaps found for the ParT trained JetClass dataset (Legge et al., 2025).

However, when examining the aggregate distribution of these scores, the overall attention score distribution closely matches the highly polarized patterns recently observed in the non-equivariant Particle Transformer (ParT) for top tagging (Legge et al., 2025). Just as in ParT, this statistical sparsity suggests the model naturally learns to isolate hard sub-jet features while aggressively discarding soft radiation. This suggests that despite the mathematical complexity of processing 16-dimensional multivectors, the geometric operations ultimately collapse into intuitive physical groupings.

Furthermore, analyzing the LGATr_slim model yields complementary insights. The slim variant maintains a similarly polarized statistical distribution (see Figure 3), but its internal routing changes. Because the full multivector space is restricted to decoupled scalars and 4-vectors, the network distributes its focus differently across its attention heads. This suggests that while the exact geometric representation alters the visual flow of information, the fundamental strategy of aggressively pruning soft background particles is intrinsic to both equivariant frameworks.

5.2 Knowledge Distillation Performance

Objective. To determine if the robust physical representations of the LGATr teacher can be compressed into a non-equivariant, low-latency MLP student using raw Cartesian inputs.

Findings. As shown in the ROC curves in Figure 5, we observe an unexpected trend: the baseline MLP trained from scratch on hard labels actually outperforms the KD-trained MLP. The KD-trained model fails to bridge the performance gap with the teacher and instead suffers a noticeable degradation in tagging performance compared to the naive baseline.

This gap is quantified in Table 3 and Table 4. The baseline student achieves an AUC of 0.9501

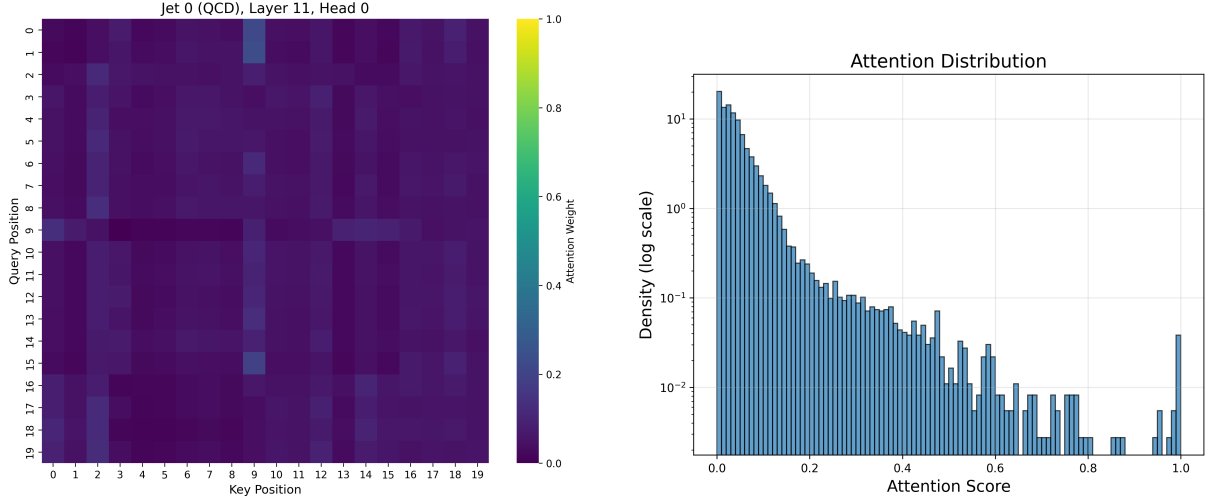


Figure 2: **LGATr Interpretation.** (Left) Attention heatmap for a sample QCD jet from LGATr. Notice the visually unstructured nature of the interactions. (Right) The overall attention score distribution. The distribution closely mirrors baseline ParT literature, aggressively filtering soft particles despite the lack of obvious visual structure in the heatmap.

and a robust background rejection of 110 at 30% signal efficiency. In stark contrast, the KD-trained student’s performance collapses, yielding an AUC of only 0.7492 and a background rejection of just 19 at the same efficiency working point.

Table 3: Test results for the baseline MLP trained from scratch on hard labels.

AUC:	0.9501
Optimal Threshold:	0.2222
Calibrated Accuracy:	0.8875
Background Rejection ($1/e_B$):	
@ 30% Signal Eff:	110
@ 50% Signal Eff:	45
@ 80% Signal Eff:	13

Table 4: Test set evaluation results for the Knowledge Distillation (KD) trained MLP.

AUC:	0.7492
Optimal Threshold:	0.0057
Calibrated Accuracy:	0.7218
Background Rejection ($1/e_B$):	
@ 30% Signal Eff:	19
@ 50% Signal Eff:	3
@ 80% Signal Eff:	2

This struggle is reflected in the training dynamics (see Figure 4). While the baseline MLP exhibits a smooth, steady loss convergence typical

of effective learning on hard labels, the KD training loss is highly volatile, noisy, and plateaus almost immediately. This optimization instability corroborates that the non-equivariant student is ill-equipped to approximate the teacher’s soft probability distributions.

We hypothesize that this discrepancy stems directly from the input representation—specifically, the “inductive bias gap” between the architectures. In successful distillation frameworks like those employing LorentzNet (Liu et al., 2023), raw inputs are explicitly converted into specialized, physics-engineered kinematic coordinates (like p_T , $\Delta\eta$, $\Delta\phi$) before being fed to the linear student.

In our LGATr pipeline, the teacher naturally ingests raw Cartesian coordinates by projecting them into equivariant multivectors. Feeding these same raw Cartesian coordinates to a non-equivariant MLP student, even with soft-label guidance, appears insufficient. The linear student lacks the inductive bias to derive relativistic invariants from raw spatial coordinates. The teacher’s softened output logits alone cannot compensate for the absence of engineered features at the input level, causing the KD optimization to flounder.

5.3 Discussion

Our two experimental tracks together yield a clear picture. The interpretability analysis reveals that while LGATr successfully prunes background radiation—matching explicitly biased models like

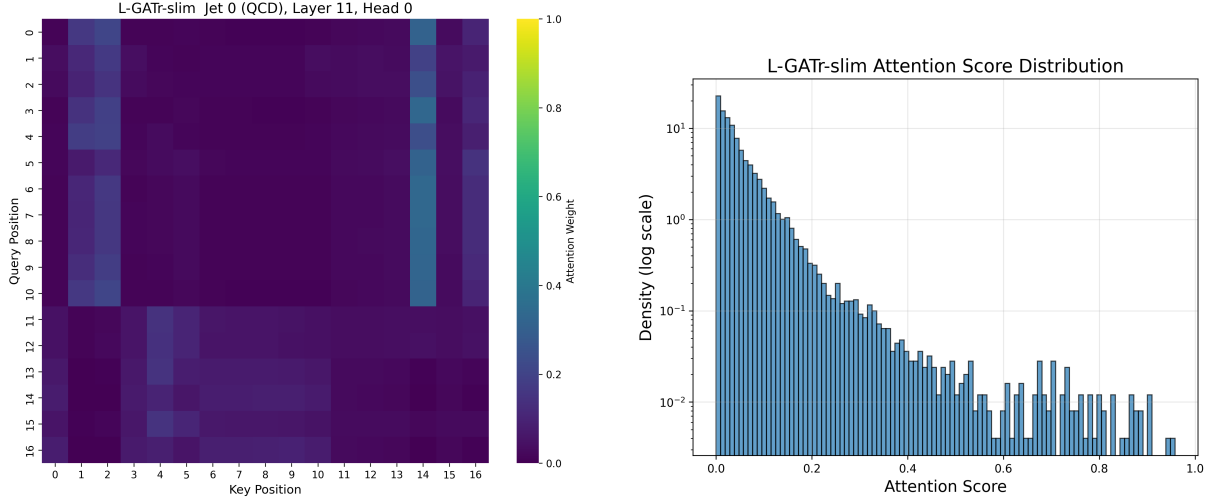


Figure 3: **LGATr_slim Interpretation.** (Left) Attention heatmap for a sample QCD jet from the restricted LGATr_slim variant. (Right) The attention score distribution, demonstrating that the statistical sparsity and filtering mechanism remains robust even when full multivector operations are ablated.

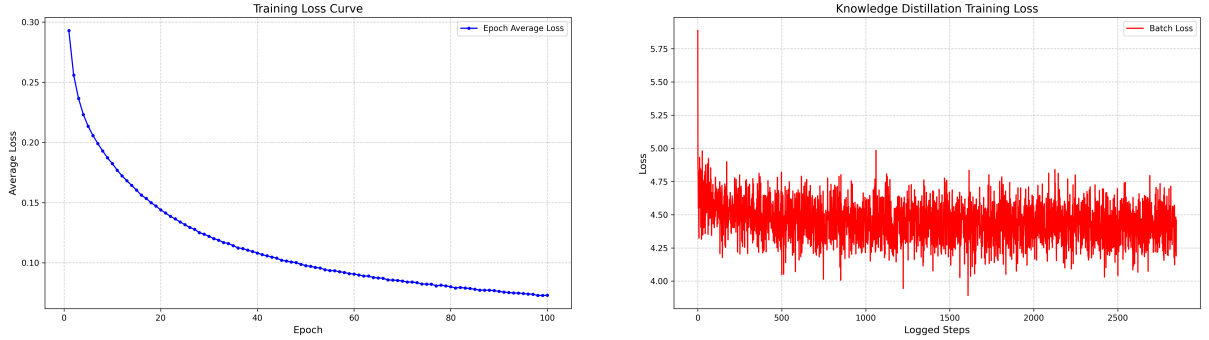


Figure 4: **Training Loss Dynamics.** (Left) The smooth, epoch-averaged training loss of the baseline MLP demonstrating stable convergence. (Right) The highly volatile, batch-level training loss of the KD-trained MLP over logged steps, illustrating severe optimization instability when learning from soft labels without appropriate physical inductive biases.

ParT in its score distributions—it does so without clear visual structure. This indicates that Geometric Algebra encodes kinematic relationships in a non-trivial manner.

Transferring this geometric reasoning via Knowledge Distillation proves difficult. The underperformance of the KD-trained MLP highlights a critical limitation in model compression for physics applications: soft labels from a complex geometric teacher cannot induce physical symmetry in a linear student if the input representation is unsuited for the student’s architecture.

This underscores the necessity of explicit feature engineering when distilling equivariant models for fast inference at the LHC. While the teacher can operate elegantly on raw Cartesians, edge-deployed students must be given physically invariant features to serve as a stepping stone.

6 Conclusion

In this paper, we presented a dual-track study exploring the transparency and efficiency of Lorentz Geometric Algebra Transformers (LGATr) for jet tagging. By applying mechanistic interpretability techniques, we demonstrated that LGATr develops visually complex but statistically sparse attention patterns whose score distributions closely mirror those of ParT, effectively learning to isolate physically relevant sub-jet structures.

In our knowledge distillation track, we attempted to transfer LGATr’s representations to a low-latency MLP student suitable for hardware triggers. We found that a lightweight MLP student trained from scratch actually outperformed the student distilled from the LGATr teacher. We identified the pipeline’s reliance on raw Cartesian coor-

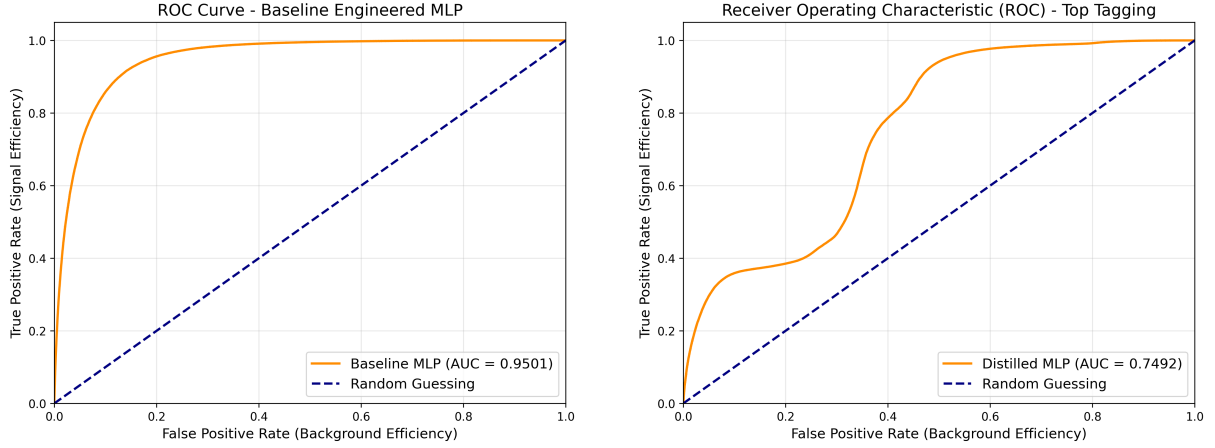


Figure 5: **Knowledge Distillation vs. Baseline Performance.** (Left) Performance of the baseline MLP student trained entirely from scratch using hard labels. (Right) Performance of the identical MLP architecture trained via Knowledge Distillation (KD). Contrary to expectations, the baseline outperforms the distilled model, likely due to the student’s inability to process raw Cartesian features effectively without inherent physical inductive biases.

dinates as the primary bottleneck, concluding that soft labels alone cannot impart geometric inductive biases to simple architectures without the aid of engineered kinematic features. These results provide a valuable lesson for future deployments of fast-inference models in microsecond-latency environments like LHC triggers.

Limitations

- **Interpretability Scope:** Our current interpretability analysis is primarily observational. Although attention score distributions show correlation with kinematic variables, attention does not always equal explanation. Future work should incorporate linear probes, layer-by-layer equivariance checks, and analyze the individual contribution of each geometric grade to verify strictly causal routing.
- **Distillation Bottleneck:** We hypothesize that the KD failure stems from feeding raw Cartesian coordinates to an MLP, but a significant limitation of this work is the absence of targeted ablation studies to empirically confirm this. To definitively isolate this failure mode, our methodology currently lacks three necessary control experiments: (1) evaluating a baseline MLP trained from scratch strictly on raw Cartesians; (2) testing a hybrid pipeline where the student receives engineered kinematic features (p_T, η, ϕ) while the teacher uses Cartesians; and (3) employing feature-based distillation to match intermediate geometric represen-

tations rather than relying solely on final output logits.

- **Student Architecture Limits:** The MLP relies on padded and flattened inputs. Distilling into a lightweight permutation-invariant architecture (like DeepSets) might yield better baseline results before distillation is applied.
- **Computational Limits & Generalization:** Due to compute limitations, our experiments were constrained to the TopTagging dataset. Future studies should validate these interpretability and KD scaling findings on the larger, more complex JetClass dataset, which is the benchmark standard in recent literature.

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